



WOLLO University

Department of software engineering

Exit multiple choice questions for

Fundamental software engineering

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1. Software is defined as _____
- a) set of programs, documentation & configuration of data
 - b) set of programs
 - c) documentation and configuration of data
 - d) None of the mentioned

[View Answer](#)

Answer: a

Explanation: Software is a collection of programmes; it also includes documentation and data setup to enable the programmes to function. Microsoft windows, excel, word, powerpoint, etc. are a few examples of software.

2. What is Software Engineering?
- a) Designing a software
 - b) Testing a software
 - c) Application of engineering principles to the design a software
 - d) None of the above

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Answer: c

Explanation: Software engineering is the application of engineering principles to the design, development, and support of software and it helps to solve the challenges of low-quality software projects.

3. Who is the father of Software Engineering?
- a) Margaret Hamilton
 - b) Watts S. Humphrey
 - c) Alan Turing
 - d) Boris Beizer

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Answer: b

Explanation: Watts S. Humphrey created the Software Process Program at Carnegie Mellon University's Institute (SEI) in the 1980s, and served as its director from 1986 through the early 1990s. This program was designed to help participants understand and manage the software development process.

4. What are the features of Software Code?
- a) Simplicity
 - b) Accessibility
 - c) Modularity
 - d) All of the above

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Answer: c

Explanation: Software code should be written in a clear, succinct, and easy-to-understand way. Simplicity should be preserved in the program code's organization, implementation, and design. These codes should be constructed in such a way that software components (such as files and functions) are readily available. The software may be broken down into numerous parts to make it easier to comprehend and troubleshoot.

5. _____ is a software development activity that is not a part of software processes.
- a) Validation
 - b) Specification

- c) Development
- d) Dependence

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Answer: d

Explanation: A software dependency is an external independent library that can range in size from a single file to numerous files and directories arranged into packages to accomplish a specified purpose and is an attribute and not an engineering activity for process.

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6. Define Agile scrum methodology.

- a) project management that emphasizes incremental progress
- b) project management that emphasizes decremental progress
- c) project management that emphasizes neutral progress
- d) project management that emphasizes no progress

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Answer: a

Explanation: Agile scrum methodology is a style of project management that emphasizes incremental progress. Each iteration is divided into two to four-week sprints, with the goal of completing the most important features first and delivering a possibly deliverable product at the end of each sprint.

7. CASE stands for

- a) Computer-Aided Software Engineering
- b) Control Aided Science and Engineering
- c) Cost Aided System Experiments
- d) None of the mentioned

[View Answer](#)

Answer: a

Explanation: The CASE tool's purpose is to make the work of software development and maintenance easier and more reliable.

8. _____ is defined as the process of generating analysis and designing documents?

- a) Re-engineering
- b) Reverse engineering
- c) Software re-engineering
- d) Science and engineering

[View Answer](#)

Answer: b

Explanation: The method of reverse engineering is used to uncover difficult, unknown, and hidden information about a software system.

9. The activity that distributes estimated effort across the planned project duration by allocating the effort to specific software developing tasks is _____

- a) Project scheduling
- b) Detailed schedule
- c) Macroscopic schedule
- d) None of the mentioned

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Answer: a

Explanation: Software project scheduling is the process of allocating anticipated effort to

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specific software developing activities and distributing it across the project's intended length. A macroscopic schedule is created in the early phases of project planning.

10. What is a Functional Requirement?

- a) specifies the tasks the program must complete
- b) specifies the tasks the program should not complete
- c) specifies the tasks the program must not work
- d) All of the mentioned

[View Answer](#)

Answer: a

Explanation: A functional requirement is a specification that describes a system or one of its components. It specifies the tasks the program must complete. A function is made up of three parts: inputs, behavior, and outputs.

11. Why do bugs and failures occur in software?

- a) Because of Developers
- b) Because of companies
- c) Because of both companies and Developers
- d) None of the mentioned

[View Answer](#)

Answer: c

Explanation: Software businesses are in charge of establishing policies and creating a working environment for software development, therefore they are a component of the software development process. Bugs from developers are nothing new.

12. Attributes of good software is _____

- a) Development
- b) Maintainability & functionality
- c) Functionality
- d) Maintainability

[View Answer](#)

Answer: b

Explanation: Good software should provide the functionality and maintainability that are necessary. Software development is a must, not an option. Along with this software also additional attributes like usability, efficiency, reliability, accuracy, robustness, integrity, etc.

13. The Cleanroom philosophy was proposed by _____

- a) Linger
- b) Mills
- c) Dyer
- d) All of the Mentioned

[View Answer](#)

Answer: d

Explanation: Mills, Dyer, and Linger initially suggested the Cleanroom concept in the 1980s. The main goal of the philosophy of cleanroom philosophy is to develop software with zero defects.

14. What does SDLC stands for?

- a) System Design Life Cycle
- b) Software Design Life Cycle
- c) Software Development Life Cycle

d) System Development Life cycle

[View Answer](#)

Answer: c

Explanation: The Software Development Life Cycle (SDLC) is a method for designing, developing, and testing high-quality software. The software developed to meet or exceed customer expectations must have an SDLC designed to complete the software on time and on budget.

15. Who proposed the spiral model?

a) Barry Boehm

b) Pressman

c) Royce

d) IBM

[View Answer](#)

Answer: a

Explanation: The spiral model by Boehm, is a software process model that combines prototyping's iterative characteristic with the linear sequential model's regulated and systematic elements. It implements the capability of quick production of new software versions.

16. _____ is not among the eight principles followed by the Software Code of Ethics and Professional Practice.

a) PRODUCT

b) ENVIRONMENT

c) PUBLIC

d) PROFESSION

[View Answer](#)

Answer: b

Explanation: The rest are software ethical provisions; the environment does not focus on individual clauses or their importance in relation to the topic. The eight principles that should be followed by the Software code is:

1. PUBLIC

2. CLIENT AND EMPLOYER

3. PRODUCT

4. JUDGMENT

5. MANAGEMENT

6. PROFESSION

7. COLLEAGUES

8. SELF

17. Which of the following are CASE tools?

a) Central Repository

b) Integrated Case Tools

c) Upper Case Tools

d) All of the mentioned

[View Answer](#)

Answer: d

Explanation: CASE tools can be divided into Central Repository, Integrated Case Tools, Upper Case Tools, and Lower Case Tools based on their use at a particular SDLC stage.

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18. _____ suits the Manifesto for Agile Software Development.

- a) Customer collaboration
- b) Individuals and interactions
- c) Working software
- d) All of the mentioned

[View Answer](#)

Answer: d

Explanation: None.

19. Software patch is defined as _____

- a) Daily or routine Fix
- b) Required or Critical Fix
- c) Emergency Fix
- d) None of the mentioned

[View Answer](#)

20. _____ software development team has no permanent leader.

- a) Controlled Centralized (CC)
- b) Controlled decentralized (CD)
- c) Democratic decentralized (DD)
- d) None of the mentioned

[View Answer](#)

Answer: c

Explanation: Team members Communicate among themselves horizontally. There is no permanent leader in the team and a team of software engineers coordinates among themselves temporarily to perform the task.

21. Regardless of application area, project size, or complexity, software development work may be divided into three generic phases: the _____ phase, which focuses on *what*, the _____ phase, which focuses on *how*, and the _____ phase, which focuses on *change*.

- i. support
- ii. development
- iii. definition

- a) iii, ii, i
- b) iii, i, ii
- c) i, ii, iii
- d) ii, i, iii

[View Answer](#)

Answer: a

Explanation: Irrespective of the application area, project size, or complexity, software development work may be divided into three generic phases: the definition phase, which focuses on *what*, the development phase, which focuses on *how*, and the support phase, which focuses on *change*.

22. _____ is not a fundamental activity for software processes in software development.

- a) Evolution
- b) Design and implementation
- c) Validation
- d) Verification

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Answer: d

Explanation: Implementation and testing activities account for software verification. Software Verification is a process that ensures that the software developed is accurate and meets the user's expectations.

23. What are agile manifesto principles?

- a) Customer satisfaction
- b) Face-to-face communication within a development team
- c) Changes in requirements are welcome
- d) All of the mentioned

[View Answer](#)

Answer: d

Explanation: Principles of the agile manifesto are:

- i) Customer satisfaction is a priority.
- ii) Changes in requirements are welcome, if they occur late in the development process.
- iii) Deliver working software on a regular basis.
- iv) Business people and developers must connect on a daily basis through the project.
- v) Support people who are interested and passionate about developing.
- vi) Face-to-face communication within a development team.
- vii) Working software indicates progress.
- viii) Sustainable development is aided by agile processes.
- ix) A constant focus on technical excellence and smart design improve agility.
- x) Simplicity, or the art of minimizing the amount of effort that isn't done, is critical.
- xi) Self-organizing teams
- xii) The team reflects on how to become more effective at regular intervals, then selects its behavior accordingly.

24. Faster delivery is possible with CBSE.

- a) False
- b) True

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Answer: b

Explanation: They build more dependable systems at a faster rate because they use previously validated components. The software developed by CBSE uses reusable components.

25. Who proposed Function Points?

- a) Albrecht
- b) Jacobson
- c) Boehm
- d) Booch

[View Answer](#)

Answer: a

Explanation: Since Albrecht presented functional points in 1979, hundreds of books and articles have been produced on the subject.

26. _____ is a software development life cycle model that is chosen if the development team has less experience on similar projects.

- a) Iterative Enhancement Model
- b) RAD
- c) Spiral

d) Waterfall

[View Answer](#)

Answer: c

Explanation: For many applications, relying on risk assessment/analysis gives more freedom than is necessary, overcoming the requirements of less experienced developers.

27. Agile Software Development is based on which of the following type?

a) Iterative Development

b) Incremental Development

c) Both Incremental and Iterative Development

d) Linear Development

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Answer: c

Explanation: The software is built in increments, with the client stating the criteria to be included in each increment, and the top goal is to please the customer by delivering valuable software early and frequently. They're iterative because they work on one iteration before moving on to the next.

28. _____ is a software developing team has a defined leader who coordinates specific tasks and secondary leaders that have responsibility for sub tasks.

a) Democratic decentralized (DD)

b) Controlled centralized (CC)

c) Controlled decentralized (CD)

d) None of the mentioned

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Answer: c

Explanation: Problem-solving is still a group activity, but the team leader divides the execution of solutions into subgroups and it is known as a Controlled decentralized team. In Democratic decentralized, there is no permanent leader in the team and a team of software developers coordinates among themselves temporarily to perform the task.

29. 4GT Model is a set of _____

a) Programs

b) CASE Tools

c) Software tools

d) None of the mentioned

[View Answer](#)

Answer: c

Explanation: 4GT is a collection of software tools that allow a software developer to describe high-level features and have source code created automatically based on those requirements.

30. Engineers developing software should not

a) be dependent on their colleagues

b) maintain integrity and independence in their professional judgment

c) not knowingly accept work that is outside your competence

d) not use your technical skills to misuse other people's computers

[View Answer](#)

31. _____ is not suitable for accommodating any change?

a) RAD Model

- b) Waterfall Model
- c) Build & Fix Model
- d) Prototyping Model

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Answer: b

Explanation: Real-world projects seldom follow the Waterfall Model's proposed sequential sequence. A Sequential model is an example of a waterfall model. The software development activity is split into several phases in this paradigm, with each phase consisting of a sequence of activities and having different goals.

32. The model which has a major disadvantage in terms of the coding phase of a software life cycle model is _____

- a) Rad Model
- b) Spiral Model
- c) 4GT Model
- d) Waterfall Model

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Answer: c

Explanation: Since the coding phase is eliminated in 4GT Model, more expertise is required for analysis, design and testing activities.

33. Adaptive Software Development(ASD) has which of the following three framework activities?

- a) speculation, collaboration, learning
- b) analysis, design, coding
- c) requirements gathering, adaptive cycle planning, iterative development
- d) all of the mentioned

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Answer: a

Explanation: None.

34. Which of the following is not a project factor that should be considered when planning the structure of software developing teams?

- a) The rigidity of the delivery date
- b) The degree of sociability required for the project
- c) High frustration caused by personal, business, or technological factors that causes friction among team members
- d) The difficulty of the problem to be solved

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Answer: b

Explanation: Development is irrelevant to the social quotient.

35. What is the full form of the "COCOMO" model?

- a) Cost Constructive Estimation Model
- b) Constructive Cost Estimation Model
- c) Constructive Case Estimation Model
- d) Constructive Cost Estimating Model

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Answer: b

Explanation: The Cocomo (Constructive Cost Model) regression model is based on the number of lines of code. It is a procedural cost estimate model that is frequently used as

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a method of accurately estimating size, effort, cost, duration, and quality for a project completion.

36. Which one of the following is not a software process quality?

- a) Visibility
- b) Timeliness
- c) Productivity
- d) Portability

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Answer: d

Explanation: Portability is a software product quality which means software can run on different hardware platforms or environments.

37. Cleanroom software development process complies with the operational analysis principles by using a method called known as

- a) referential transparency
- b) degenerative error correction
- c) box structure specification
- d) none of the mentioned

[View Answer](#)

Answer: c

Explanation: Box structures are descriptions of functions that exhibit properties essential for effective system specification and design.

38. What is system software?

- a) computer program
- b) Testing
- c) AI
- d) IOT

[View Answer](#)

Answer: a

Explanation: System software is a sort of computer program that manages the hardware and applications on a computer. They are of three types: This is the operating system. This is a language processor. Software that is useful.

39. Quality Management is known as _____

- a) SQI
- b) SQA
- c) SQM
- d) SQA and SQM

[View Answer](#)

Answer: b

Explanation: Software quality assurance (SQA) is another name for quality management, which is an umbrella activity that is used throughout the software development process.

40. _____ is the definition of objects in the database that leads directly to a standard approach for the creation of software documentation.

- a) Data/data integration
- b) Information sharing
- c) Document standardization
- d) Data integrity

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Answer: c

Explanation: In a software project, documentation standards are critical since papers are the only concrete means to describe the program and the development process.

Standardized papers should be easy to read and comprehend since they have a uniform look, structure, and quality.

41. _____ is an indirect measure of software development process.

- a) Cost
- b) Effort Applied
- c) Efficiency
- d) All of the mentioned

[View Answer](#)

Answer: c

Explanation: Efficiency is an indirect measure. Indirect measures also include products like maintainability, quality, functionality, complexity, reliability, and many more.

42. According to an IBM research, “31% of projects are abandoned before they are completed, 53% exceed their cost projections by an average of 189 percent, and 94 projects are restarted for every 100 projects.” What is the significance of these figures?

- a) Lack of software ethics and understanding
- b) Management issues in the company
- c) Lack of adequate training
- d) All of the mentioned

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Answer: c

Explanation: Lack of software ethics and knowledge, as well as corporate management difficulties, are all aspects of Software Development, therefore this is a possibility. Both are due to a lack of sufficient training.

43. Which of the following document contains the user system requirements?

- a) SRD
- b) DDD
- c) SDD
- d) SRS

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Answer: d

Explanation: A software requirements specification (SRS) is a detailed explanation of how a system should behave before it is built. It may also include a collection of use cases that explain how users will interact with the programme.

44. _____ specification is also known as SRS document.

- a) white-box
- b) grey-box
- c) black-box
- d) none of the mentioned

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Answer: c

Explanation: The system is regarded as a black box, with no knowledge of its underlying workings, and just its observable exterior (input/output) behavior described.

45. Which of the following is not a part of Software evolution?

- a) Re-engineering activities
- b) Maintenance activities

- c) Development activities
- d) Negotiating with client

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Answer: d

Explanation: The study and management of the process of making changes to software through time is referred to as software evolution. As a result, the remaining three alternatives are available.

46. _____ is a Strategy to achieve Software diversity.

- a) Explicit specification of different algorithms
- b) Different programming languages
- c) Different design methods and tools
- d) All of the mentioned

[View Answer](#)

Answer: d

Explanation: Diversity refers to the ability to deliver the same functionality in a variety of ways so that essential components of a dependable system do not fail in the same way. Because we all have varied life experiences, backgrounds, and expertise, adding variety to the problem-solving process is essential because it allows us to come up with new ideas and methods.

47. In which step of SDLC actual programming of software code is done?

- a) Development and Documentation
- b) Maintenance and Evaluation
- c) Design
- d) Analysis

[View Answer](#)

Answer: a

Explanation: The documentation explains the functions of the final product. The developer must discover adequate knowledge in the technical documentation to begin coding.

48. Software Debugging is known as _____

- a) identifying the task to be computerized
- b) creating program code
- c) creating the algorithm
- d) finding and correcting errors in the program code

[View Answer](#)

49. The word which describes the importance of software design is?

- a) Complexity
- b) Quality
- c) Efficiency
- d) Accuracy

[View Answer](#)

Answer: b

Explanation: The degree to which software complies with or adheres to a particular design based on functional requirements or specifications is referred to as functional quality.

50. The incorrect activity among the following for the configuration management of a software system is _____

- a) Version management

- b) System management
- c) Change management
- d) Internship management

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Answer: d

Explanation: The policies and methods for configuration management describe how to track and process proposed system changes, as well as how to select which system components to modify, how to manage various versions of the system and its components, and how to disseminate changes to customers.

1) What is the first step in the software development lifecycle?

- a. System Design
- b. Coding
- c. System Testing
- d. Preliminary Investigation and Analysis

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Answer: d) Preliminary Investigation and Analysis

Explanation: The first phase of the software development life cycle is the requirement analysis, in which the software engineer analyzes the software requirement based on the user's outlook.

2) What does the study of an existing system refer to?

- a. Details of DFD
- b. Feasibility Study
- c. System Analysis
- d. System Planning

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Answer: c) System Analysis

Explanation: The study of an existing system refers to system analysis.

3) Which of the following is involved in the system planning and designing phase of the Software Development Life Cycle (SDLC)?

- a. Sizing
- b. Parallel run
- c. Specification freeze
- d. All of the above

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Answer: d) All of the above

Explanation: None.

4) What does RAD stand for?

- a. Rapid Application Document
- b. Rapid Application Development
- c. Relative Application Development
- d. None of the above

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Answer: b) Rapid Application Development

Explanation: RAD stands for Rapid Application Development is categorized as an agile development method, which is meant to accomplish a quick turnaround and high-end outcomes.

5) Which of the following prototypes does not associated with Prototyping Model?

- a. Domain Prototype
- b. Vertical Prototype
- c. Horizontal Prototype
- d. Diagonal Prototype

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Answer: d) Diagonal Prototype

Explanation: There is nothing as such called Diagonal Prototype; however, the rest other options do make sense.

6) The major drawback of RAD model is _____.

- a. It requires highly skilled developers/designers.
- b. It necessitates customer feedbacks.
- c. It increases the component reusability.
- d. Both (a) & (c)

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Answer: d) Both (a) & (c)

Explanation: The client may create an unrealistic product vision, which may result in a team over or under-develop functionality. Besides, it is not easy to hire highly specialized & skilled developers/designers.

7) Which of the following does not relate to Evolutionary Process Model?

- a. Incremental Model
- b. Concurrent Development Model
- c. WINWIN Spiral Model
- d. All of the above

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Answer: d) All of the above

Explanation: The evolutionary process model is designed for rapid software development, which starts from scratch with raw specifications and modifies as per the user requirement.

8) What is the major drawback of the Spiral Model?

- a. Higher amount of risk analysis
- b. Doesn't work well for smaller projects
- c. Additional functionalities are added later on
- d. Strong approval and documentation control

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Answer: b) Doesn't work well for smaller projects

Explanation: A spiral model is an incremental approach, which is formed as a combination of the waterfall model and prototyping model. The major drawbacks of the Spiral model are as follows:

- Expensive
- Doesn't work well for smaller projects
- Risk analysis requires highly skilled experts.

9) Model selection is based on _____.

- a. Requirements
- b. Development team & users
- c. Project type & associated risk
- d. All of the above

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Answer: d) All of the above

Explanation: Model selection is based on requirements, a team of developers, users as well as the risk involved in developing a project.

10) Which of the following option is correct?

- a. The prototyping model facilitates the reusability of components.
- b. RAD Model facilitates reusability of components
- c. Both RAD & Prototyping Model facilitates reusability of components

d. None

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Answer: c) Both RAD & Prototyping Model facilitates reusability of components

Explanation: None.

11) Which of the following models doesn't necessitate defining requirements at the earliest in the lifecycle?

- a. RAD & Waterfall
- b. Prototyping & Waterfall
- c. Spiral & Prototyping
- d. Spiral & RAD

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Answer: c) Spiral & Prototyping

Explanation: In the Prototyping model, the first phase is the requirement analysis phase, which involves Brainstorming, QFD, and FAST, whereas the Spiral model encompasses customer communication activities such as defining objectives.

12) When the user participation isn't involved, which of the following models will not result in the desired output?

- a. Prototyping & Waterfall
- b. Prototyping & RAD
- c. Prototyping & Spiral
- d. RAD & Spiral

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Answer: b) Prototyping & RAD

Explanation: The Prototyping model necessitates user participation as when a prototype is created or designed, whereas, in the case of RAD, user participation is required in all four phases.

13) Which of the following model will be preferred by a company that is planning to deploy an advanced version of the existing software in the market?

- a. Spiral
- b. Iterative Enhancement
- c. RAD
- d. Both (b) and (c)

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Answer: d) Both (b) and (c)

Explanation: None.

14) Arrange the following activities for making a software product by utilizing 4GT.

- I. Design strategy
- II. Transformation into product
- III. Implementation
- IV. Requirement gathering

- a. 4, 1, 3, 2
- b. 4, 3, 2, 1
- c. 1, 2, 3, 4
- d. 1, 4, 2, 3

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Answer: a) 4, 1, 3, 2

Explanation: The sequence of activities mentioned in option c represents the Fourth Generation Techniques(4GT) Model.

15) Which of the following is an example of Black Box and Functional Processing?

- a. First Generation Language
- b. Second Generation Language
- c. Third Generation Language
- d. Fourth Generation Language

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Answer: d) Fourth Generation Language

Explanation: Functional processing or testing is commonly known as black-box testing, where the existing contents are almost unknown, so anything could be demonstrated as a black box. Hence the functionality of a black box can be defined in terms of inputs and outputs.

16) _____ is identified as fourth generation language.

- a. Unix shell
- b. C++
- c. COBOL
- d. FORTRAN

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Answer: a) Unix shell

Explanation: Fourth-generation language is an advancement of third-generation programming language.

17) The productivity of a software engineer can be reduced by using a 4GT.

- a. True
- b. False

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Answer: b) False

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Explanation: The fourth-generation programming languages are more programmer-friendly and helpful in augmenting the program's efficiency as it uses English words and phrases. It increases the programmer's productivity by enabling them to engage in the software development life cycle.

18) Which of the following is the main advantage of deploying a 4GT model for producing small-scale products, programs, and applications?

- a. The productivity of software engineers is improved.
- b. The time required for developing software is reduced.
- c. CASE tools and code generators help the 4GT model by providing a credible solution to their problems.
- d. None of the above.

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Answer: b) Time required for developing software is reduced.

Explanation: Since automated coding is done using CASE tools & code generators, proponents claim a dramatic reduction in software development time.

19) Which of the following model has a major downfall to a software development life cycle in terms of the coding phase?

- a. 4GT Model
- b. Waterfall Model
- c. RAD Model
- d. Spiral Model

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Answer: a) 4GT Model

Explanation: Much more expertise is needed in the 4GT model for analyzing, designing, and testing activities as it eliminates the coding phase.

20) Which of the following falls under the category of software products?

- a. Firmware, CAD
- b. Embedded, CAM
- c. Customized, Generic
- d. CAD, Embedded

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21) Software maintenance costs are expensive in contrast to software development.

- a. True
- b. False

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Answer: a) True

Explanation: For systems with long life, software maintenance cost is higher than that of the software development cost.

22) Which of the following activities of the generic process framework delivers a feedback report?

- a. Deployment
- b. Planning
- c. Modeling
- d. Construction

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23) Which of the following refers to internal software equality?

- a. Scalability
- b. Reusability
- c. Reliability
- d. Usability

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Answer: b) Reusability

Explanation: None.

24) RUP is abbreviated as _____, invented by a division of _____.

- a. Rational Unified Process, IBM
- b. Rational Unified Program, IBM
- c. Rational Unified Process, Infosys
- d. Rational Unified Program, Infosys

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Answer: a) Rational Unified Process, IBM

Explanation: None.

25) The RUP can be defined based on three main perspectives, i.e., static, dynamic & practice. What is the purpose of static perspective?

- a. It recommends some good practices that be included while carrying out the process.
- b. It displays all the enacted process activities.
- c. It portrays the phases that the model has entered over a specific time.
- d. All of the above.

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Answer: b) It displays all the enacted process activities.

Explanation: None.

26) Which one of the following activities is not recommended for software processes in software engineering?

- a. Software Evolution

- b. Software Verification
- c. Software Testing & Validation
- d. Software designing

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Answer: b) Software Verification

Explanation: Software verification is mainly considered for implementing and testing activities.

27) Arrange the following activities to form a general software engineering process model.

I. Manufacture

II. Maintain

III. Test

IV. Install

V. Design

VI. Specification

- a. 6, 5, 1, 3, 4, 2
- b. 1, 2, 4, 3, 6, 5
- c. 6, 1, 4, 2, 3, 5
- d. 1, 6, 5, 2, 3, 4

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Answer: a) 6, 5, 1, 3, 4, 2

Explanation: None.

28) The agile software development model is built based on _____.

- a. Linear Development
- b. Incremental Development
- c. Iterative Development
- d. Both Incremental and Iterative Development

Hide Answer Workspace

Answer: d) Both Incremental and Iterative Development

Explanation: In agile software development, the software is usually developed in increments with the customer specifying necessary requirements in each increment. Here the main aim of a developer is to meet customer satisfaction at the earliest by providing a valuable software product. It is known as iterative because it chooses to work on one particular increment and make the improvement in other iteration.

29) On what basis is plan-driven development different from that of the software development process?

- a. Based on the iterations that occurred within the activities.
- b. Based on the output, which is derived after negotiating in the software development process.
- c. Based on the interleaved specification, design, testing, and implementation activities.
- d. All of the above

Hide Answer Workspace

Answer: a) Based on the iterations that occurred within the activities.

Explanation: In software engineering, a plan-driven approach is based on several development stages that result in the output at each of these stages, which are already planned in advance.

30) Which of the following activities is not applicable to agile software development?

- a. Producing only the essential work products.
- b. Utilizing the strategy of incremental product delivery.

- c. Abolishing the project planning and testing.
- d. All of the above

Hide Answer Workspace

Answer: c) Abolishing the project planning and testing.

Explanation: Testing can never be avoided as it plays a significant role in the software development life cycle.

31) Which of the following framework activities are carried out in Adaptive Software Development (ASD)?

- a. Assumption, Association, Learning
- b. The investigation, Strategy, Coding
- c. Requirements gathering, Adaptive cycle planning, Iterative development
- d. All of the above

Hide Answer Workspace

Answer: a) Assumption, Association, Learning

Explanation: An Adaptive Software Development approach is an empirical process control model, which is based on observations. It continuously adjusts the design of the solution and the process involved to create the solution based on observation throughout the project.

32) The _____ model helps in representing the system's dynamic behavior.

- a. Object Model
- b. Context Model
- c. Behavioral Model
- d. Data Model

Hide Answer Workspace

Answer: c) Behavioral Model

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Explanation: To represent the system's dynamic behavior, Behavioral models are used. The behavioral model is modeled from the perspective of the system's processed data or the events that stimulate responses from a system.

33) Model-driven engineering is nothing but a theoretical concept. It can never be transmuted into a working/executable code.

- a. True
- b. False

Hide Answer Workspace

Answer: b) False

Explanation: Model-driven engineering is a software development approach that represents the system as a set model, which can be easily altered into an executable code.

34) The _____ and _____ are the two major dimensions encompassed in the Spiral model.

- a. Diagonal, Perpendicular
- b. Perpendicular, Radial
- c. Angular, diagonal
- d. Radial, Angular

Hide Answer Workspace

Answer: d) Radial, Angular

Explanation: The cumulative cost is represented by the radial dimension, whereas the angular dimension represents the progress made in the completion of each consecutive cycle. Each loop in the spiral model depicts the phase.

35) Which of the following technique is involved in certifying the sustained development of legacy systems?

- a. Reengineering

- b. Forward engineering
- c. Reverse engineering
- d. Reverse engineering and Reengineering

Hide Answer Workspace

Answer: d) Reverse engineering and Reengineering

Explanation: The process involved in certifying the sustained development of legacy systems relies on Reverse engineering and Reengineering.

36) An erroneous system state that results in an unexpected system behavior is acknowledged as?

- a. System failure
- b. Human error or mistake
- c. System error
- d. System fault

Hide Answer Workspace

Answer: c) System error

Explanation: None.

37) What is the name of the approach that follows step-by-step instructions for solving a problem?

- a. An Algorithm
- b. A Plan
- c. A List
- d. Sequential Structure

Hide Answer Workspace

Answer: a) An Algorithm

Explanation: An algorithm is a precise step-by-step technique that takes an input instance (of the given problem) as input(s) and produces output for the problem instance.

38) Which of the following word correctly summarized the importance of software design?

- a. Quality
- b. Complexity
- c. Efficiency
- d. Accuracy

Hide Answer Workspace

Answer: a) Quality

Explanation: The quality of a functional software functional depicts how well it can conform with or adapts to a given design based on functional requirements or specifications.

39) Which of the following methodology results in SA/SD features?

- a. Gane and Sarson methodology
- b. DeMarco and Yourdon methodology
- c. Constantine and Yourdon methodology
- d. All of the above

Hide Answer Workspace

Answer: d) All of the above

Explanation: None.

40) _____ is not considered as an activity of Structured Analysis (SA).

- a. Transformation of a textual problem description into a graphic model

- b. Functional decomposition
- c. All the functions represented in the DFD are mapped to a module structure
- d. d) All of the mentioned

Hide Answer Workspace

Answer: c) All the functions represented in the DFD are mapped to a module structure

Explanation: The module structure depicts the architecture of the software.

41) What does a directed arc or line signify?

- a. Data Flow
- b. Data Process
- c. Data Stores
- d. None of the above

Hide Answer Workspace

Answer: a) Data Flow

Explanation: In the business information system, the data flow diagram is used to epitomize the flow of the data.

42) What does a data store symbol in the Data Flow Diagram signify?

- a. Logical File
- b. Physical File
- c. Data Structure
- d. All of the above

Hide Answer Workspace

Answer: d) All of the above

Explanation: A logical file can be a data structure or physical file on the disk or vice versa.

43) _____ is not a direct measure of SE process.

- a. Effort
- b. Cost
- c. Efficiency
- d. All of the above

Hide Answer Workspace

Answer: c) Efficiency

Explanation: Efficiency can never be measured as it is an indirect measure of the SE process.

44) What is the main task of project indicators?

- a. To evaluate the ongoing project's status and track possible risks.
- b. To evaluate the ongoing project's status.
- c. To track potential risks.
- d. None of the above

Hide Answer Workspace

Answer: a) To evaluate the ongoing project's status and track possible risks.

Explanation: None.

45) What is the main intent of project metrics?

- a. For strategic purposes
- b. To minimize the development schedule.
- c. To evaluate the ongoing project's quality on a daily basis
- d. To minimize the development schedule and evaluate the ongoing project's quality on a daily basis

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Hide Answer Workspace

Answer: d) To minimize the development schedule and evaluate the ongoing project's quality on a daily basis

Explanation: A project metric refers to a quantifiable measure of the degree to which a system, component, or process owns a certain trait.

46) Name the graphical practice that depicts the meaningful changes that occurred in metrics data.

- a. Function point analysis
- b. Control Chart
- c. DRE (Defect Removal Efficiency)
- d. None of the above

Hide Answer Workspace

Answer: c) Control Chart

Explanation: A control chart is a method used to keep track of the variability of traits of interest.

47) Which parameters are essentially used while computing the software development cost?

- a. Hardware and Software Costs
- b. Effort Costs
- c. Travel and Training Costs
- d. All of the above

Hide Answer Workspace

Answer: d) All of the above

Explanation: Estimation cost works out on assessing the amount of effort required to complete each activity, followed by calculating the total cost of activities.

48) Which of the following is an incorrect activity for the configuration management of a software system?

- a. Change management
- b. System management
- c. Internship management
- d. Version management

Hide Answer Workspace

Answer: c) Internship management

Explanation: Configuration management policies and processes define how to record, and process proposed system changes, how to decide what system components to change, how to manage different versions of the system and its components, and how to distribute changes to customers.

49) The project planner examines the statement of scope and extracts all-important software functions, which is known as

- a. Planning process
- b. Decomposition
- c. Association
- d. All of the mentioned

Hide Answer Workspace

Answer: b) Decomposition

Explanation: None.

50) Which of the following is not considered as an option for achieving reliable cost and effort estimation?

- a. The ability to translate the size estimate into human effort, calendar time, and dollars
- b. Use relatively simple decomposition techniques to generate project cost and effort estimates.

- c. Base estimates on similar projects that have already been completed
- d. Use one or more empirical models for software cost and effort estimation

Hide Answer Workspace

Answer: a) The ability to translate the size estimate into human effort, calendar time, and dollars

Explanation: None.

51) Which of the following does not complement the decomposition techniques but offers a potential estimation approach for their impersonal growth?

- a. Empirical estimation models
- b. Decomposition techniques
- c. Automated estimation tools
- d. Both empirical estimation models and automated estimation tools

Hide Answer Workspace

Answer: a) Empirical estimation models

Explanation: In computer software, the estimation model uses empirically derived formulas for guessing the effort as a function of LOC or FP.

52) Which of the following is not included in the total effort cost?

- a. Costs of lunch time food
- b. Costs of support staff
- c. Costs of networking and communications
- d. Costs of air conditioning and lighting in the office space

Show Answer Workspace

53) What is developed by utilizing the historical cost function?

- a. Parkinson's Law

- b. Expert judgment
- c. Algorithmic cost modeling
- d. Estimation by analogy

Hide Answer Workspace

Answer: c) Algorithmic cost modeling

Explanation: The algorithmic cost modeling utilizes the basic regression formula with such parameters that have been derived from the data of a historical project and current as well as characteristics of a future project.

54) Which of the following model has a misconception that systems are built by utilizing reusable components, scripts, and database programs?

- a. The reuse model
- b. An early designed model
- c. An application-composition model
- d. A post architecture model

Hide Answer Workspace

Answer: c) An application-composition model

Explanation: An application-composition model evaluates the development of a prototype.

55) Which of the following is used to predict the effort as a function of LOC or FP?

- a. COCOMO
- b. FP-based estimation
- c. Both COCOMO and FP-based estimation
- d. Process-based estimation

Hide Answer Workspace

Answer: c) Both COCOMO and FP-based estimation

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Explanation: Both COCOMO and FP-based estimation is used to calculate the effort by utilizing the empirically derived formulas.

56) Once the requirements are stabilized, the basic architecture of the software can be established. Which of the following version of the COCOMO model conforms to the given statement?

- a. Application composition model
- b. Post-architecture-stage model
- c. Early design stage model
- d. All of the above

Hide Answer Workspace

Answer: a) Application composition model

Explanation: None.

57) Which of the following threatens the quality and timeliness of the produced software?

- a. Business risks
- b. Potential risks
- c. Technical risks
- d. Known risks

Hide Answer Workspace

Answer: c) Technical risks

Explanation: Technical risks detect the possible design, enactment, interface, authentication, and preservation problems.

58) Which of the following refers to the systematic attempt, which is implemented to ascertain the threats to any project plan?

- a. Performance risk

- b. Risk identification
- c. Risk projection
- d. Support risk

Hide Answer Workspace

Answer: c) Risk projection

Explanation: Once the risks are identified, the project manager takes his first step to avoid them when required and control them.

59) Which of the following standards is used by the aviation industry?

- a. CTRADO-172B
- b. RTCADO-178B
- c. RTRADO-178B
- d. CTCADO-178B

Hide Answer Workspace

Answer: b) RTCADO-178B

Explanation: RTCADO-178B is the most popular aviation standard, which has been altered to a defacto standard.

60) Third-Party Certification for software standards is based on _____.

- a. UI 1996, Second Edition
- b. UI 1998, Second Edition
- c. UI 1992, Second Edition
- d. UT 1998, Second Edition

Hide Answer Workspace

Answer: b) UI 1998, Second Edition

Explanation: None.

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If the objects focus on the problem domain, then we are concerned with

Object-oriented analysis

Object-oriented design

Object-oriented analysis and design

None

Hide

Wrong Answer

If the objects focus on the problem domain, then we are concerned with object-oriented analysis.

2.

When can white-box testing be started?

After SRS creation

After installation

After programming

After designing

Hide

Wrong Answer

White-box testing be started after programming.

3.

By whom is unit testing done?

Users

Customers

Developers

None

Hide

Wrong Answer

Unit testing is done by developers.

4.

In which of the following categories can white-box testing be classified?

Design based testing

Structural testing

Error guessing technique

None of the above

Hide

Wrong Answer

White-box testing can also be classified as structural testing.

5.

Infrastructure software is covered under which of the following categories?

Customized products

Generic products

Generic and customized products

None

Hide

Wrong Answer

Infrastructure software is covered under Generic products.

6.

What does the RAD software process stand for?

Rapid Application Development

Recent Application Development

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Relative Application Development

Rapid Application Design

Hide

Wrong Answer

RAD stands for Rapid Application Development.

7.

Identify the simplest model of SDLC?

Agile

RAD

Waterfall

Spiral

Hide

Wrong Answer

The spiral model is the simplest model of SDLC.

8.

Choose the correct option among the following. The software consists of?

Program+documentation operating procedures

Program+hardware manuals

Instruction+hardware manuals

Set of programs

Hide

Wrong Answer

Software consists of Program+documentation operating procedures.

9.

Which is the following is the most important feature of the spiral model?

Efficiency management

Time management

Risk management

Quality management

Hide

Wrong Answer

Risk management is the most important feature of the spiral model.

10.

Total phases of RAD software model is-

6

5

3

2

Hide

Wrong Answer

RAD consists of a total of 5 phases - Business modeling, Data modeling, Process modeling, application generation, and testing & turnover.

11.

Identify the incorrect phase of the prototype model.

Prototype refinement

Engineer product

Coding

Quick design

Hide

Wrong Answer

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Coding is an incorrect phase. In the prototype model, it generates only a working model.

12.

Which of the following is the first step of SDLC?

Coding

Design

Preliminary Investigation and Analysis

Testing

Hide

Wrong Answer

The first step of SDLC is Preliminary Investigation and Analysis.

13.

Choose the correct option which does not relate to the Evolutionary Process Model?

Concurrent Development Model

WINWIN Spiral Model

Incremental Model

All of the above

Hide

Wrong Answer

All of the above does not relate to the Evolutionary Process Model. The evolutionary Process Model is designed for rapid model development i.e it is developed from scratch and is modified as per user requirement.

14.

Which of the following is a fourth-generation language?

COBOL

UNIX Shell

C++

Fotran

Hide

Wrong Answer

UNIX Shell is a fourth-generation language.

15.

Select the correct option which refers to internal software equality?

Reusability

Scalability

Usability

Reliability

Hide

Wrong Answer

Reusability is referred to as internal software equality.

16.

Identify among the following which is not recommended for software processes in software engineering?

Software verification

Software testing and validation

Software designing

Software evolution

Hide

Wrong Answer

Software verification is not recommended since it is mainly used for implementing and testing.

17.

Choose the option which is not applicable to agile software development?

Producing only the essential work products

Recent Application Development

Abolishing the project planning and testing

None

Hide

Wrong Answer

Abolishing the project planning and testing is not applicable as testing and planning are essential in SDLC.

18.

Choose the correct option among the following which is involved in the system planning and designing phase of the Software Development Life Cycle (SDLC)?

Parallel run

Sizing

Specification freeze

All of the above

Hide

Wrong Answer

Parallel run, sizing, specification freeze are involved in the system planning and designing phase of the Software Development Life Cycle (SDLC).

19.

Identify the major drawback of the Spiral Model?

Does not work efficiently with small projects

High amount of risk analysis

Strong approval and documentation control

Additional functionalities are added later on

Hide

Wrong Answer

In the spiral model, the major drawback is that additional functionalities are added later on.

20.

On which of the following is model selection based upon?

Development teams and users

Requirements

Project types and the associated risks

All of the above mentioned

Hide

Wrong Answer

All of the above are necessary factors while modeling selection.

21.

Which of the following models will not result in the desired output, when the user participation isn't involved?

RAD and prototyping

Waterfall and prototyping

Prototyping and spiral

Spiral and RAD

Hide

Wrong Answer

RAD and prototyping are the models which are dependent upon the user's participation and might produce undesired output.

22.

Choose the correct option which states the correct definition of Software. It is considered to be a collection of
Programming code
documentations
associated libraries

All of the above

Hide

Wrong Answer

Software is said to be a collection of programming code, documentation, associated libraries.

23.

Identify the incorrect type of prototype of the Prototyping Model among the following.

Horizontal prototype

Diagonal prototype

Vertical prototype

None

Hide

Wrong Answer

Diagonal prototype is incorrect.

24.

For which of the following reasons is the waterfall model not suitable.

Small projects

Complex projects

Maintenance Projects

Accommodating changes

Hide

Wrong Answer

Waterfall model is not suitable for accommodating changes.

25.

Identify the correct software metrics?

Complexity Metrics

Process Metrics

Quality Metrics

All of the above

Hide

Wrong Answer

All of the above are correct software matrices,

26.

Identify the model which is also known as Verification and Validation Model?

Waterfall model

V-model

Prototype model

Evolutionary model

Hide

Wrong Answer

The V-model is also known as Verification and Validation Model.

27.

Identify the incorrect software layers.

Process

Manufacturing

Tools

Method

Hide

Wrong Answer

Manufacturing is not considered in the software layers.

28.

Which of the following is not included in the efficiency of a software product.

Processing time

licensing

Responsiveness

Memory utilization

Hide

Wrong Answer

Licensing is not included in the efficiency of a software product.

29.

Identify the two dimensions of the spiral model.

Radial, angular

Diagonal, angular

Diagonal, perpendicular

None

Hide

Wrong Answer

Radial, angular are the two types of dimensions of the spiral model.

30.

On which of the following is Agile Software Development based?

Linear development

Incremental development

Iterative development

Both incremental and iterative development

Hide

Wrong Answer

Agile Software Development based on both incremental and iterative development.

31.

FAST stands for?

Facilitated Application Specification Technique

Functional Application Specification Techniques

Fast Application Specification Technique

None

Hide

Wrong Answer

FAST stands for Facilitated Application Specification Technique.

32.

Requirement Analysis does not deal with which of the following diagrams?

Entity Relationship Diagram

Use cases

Activity diagram

State transition diagram

Hide

Wrong Answer

Requirement Analysis does not deal with the Activity diagram.

33.

The SRS document is also known as _____ specification.

Black box

White box

Grey box

None

Hide

Wrong Answer

The SRS document is also known as black box specification.

34.

Identify the correct option among the following which is not included in SRS?

Functionality

Performance

External interfaces

Design solutions

Hide

Wrong Answer

Design solutions are not included in SRS.

35.

Total different notations for Unified Modelling Language?

3

4

6

9

Hide

Wrong Answer

There are total 9 different notations for Unified Modelling Language.

36.

Identify the correct functional requirement.

Robustness

Portability

Maintainability

None

Hide

Wrong Answer

All of the above are non-functional requirements, hence D) is correct.

37.

Identify the correct measure for correctness.

Errors per KLOC

\$ per KLOC

Defects per KLOC

None

Hide

Wrong Answer

Defects per KLOC is the most common measure for correctness.

38.

Identify the fault-based testing technique.

Beta testing

Unit testing

Mutation testing

Stress testing

Hide

Wrong Answer

Mutation testing is a fault-based technique.

39.

UML diagram cannot be used to capture which of the following views?

Layer view

User view

Structural view

None

Hide

Wrong Answer

Layer view cannot be captured using UML diagram.

40.

The construction of a simpler version of a problem by ignoring details is known as

Encapsulation

Abstraction

Decomposition

Composition

Hide

Wrong Answer

The construction of a simpler version of a problem by ignoring details is known as abstraction.

41.

Which of the following terms is used often for the principle of abstraction?

Modelling

Data binding

Propelling

Manipulating

Hide

Wrong Answer

Principal of abstraction is also known as modelling.

42.

Decomposition is also popularly known as

Group and merge principle

Binary search principle

Quick sort principle

Divide and conquer principle

Hide

Wrong Answer

Decomposition is also popularly known as Divide and conquer principle.

43.

Identify the term which is not related to testing?

failure

error

Test case

Test bot

Hide

Wrong Answer

Test bot is not related to testing.

44.

Identify the incorrect testing technique.

Integration testing

Collaboration testing

System testing

Unit testing

Hide

Wrong Answer

Collaboration testing is not a testing technique.

45.

Which of the following are type of code review?

Code walkthrough

Code inspection

Both A and B

None

Hide

Wrong Answer

Both A and B are types of code review.

46.

Identify the worst type of coupling.

Data coupling

Control coupling

Stam coupling

Content coupling

Hide

Wrong Answer

In a content coupling, one module can modify the data of another module, or control flow is passed from one module to the other module. This is the worst form of coupling and should be avoided.

47.

RUP and PSP stands for?

Rational Unified Process and Personal software process

Rational Unified Process and Project software process

Real-time unified process and Project software process

Real-time unified process and Personal software process

Hide

Wrong Answer

RUP stands for Rational Unified Process and PSP stands for Personal software process.

48.

When is the 'risk analysis' In the spiral model performed?

In the first loop

Before using the spiral model

Every loop

The first and second loop

Hide

Wrong Answer

The 'risk analysis' In the spiral model is performed in every loop.

49.

Which feature of OOPs helps in code reusability?

Class

Inheritance

Abstraction

Object

Hide

Wrong Answer

Inheritance helps in code reusability.

50.

COCOMO model stand for?

Complete Cost Estimation Model

Constructive Cost Estimation Model

Common Cost Estimation Model

Comprehensive Cost Estimation Model

Hide

Wrong Answer

COCOMO stands for Constructive Cost Estimation Model.